

Welcome to the Course

Anomaly Detection

- ✔ **Video:** Detecting Anomalies
5 min
- ✔ **Video:** Detecting Anomalies in MATLAB
6 min
- ✔ **Reading:** Installing Pre-Trained Models
10 min
- ✔ **Reading:** PCB Anomaly Detection with PatchCore and FCDD
1h

Anomaly Detection Assessment

PCB Anomaly Detection with PatchCore and FCDD

Navigate to the Module 1 folder to find two live scripts detailing how to perform anomaly detection on printed circuit board (PCB) images using PatchCore and FCDD:

- DetectingPCBAnomaliesPatchCore.mlx
- DetectingPCBAnomaliesFCDD.mlx

The PatchCore anomaly detector is much more powerful than its FCDD counterpart, but it also requires a significant amount of resources to train and use. **If you do not have a GPU, we do not recommend training or using the model in DetectingPCBAnomaliesPatchCore.mlx.** Run the file using the provided results instead to better understand the steps to train a PatchCore anomaly detector. The code here is the same as that in the previous video, but with additional explanation.

By comparison, the FCDD detector is much faster and more lightweight. **Try training the model in DetectingPCBAnomaliesFCDD.mlx.** You should be able to train an anomaly detector in under an hour, even on a CPU. While the general process is similar, pay special attention to the key differences between training an FCDD and PatchCore anomaly detector. Later in this course, you will work with an FCDD model for detecting anomalies in endoscopy images.

Go to next item ✓ Completed

👍 Like 🗨 Dislike 🚩 Report an issue

